Statistics (Review)

NOU

Basics of Statistics, Jarkko Isotalo

Introduction

- *"Statistics consists of a body of methods for collecting and analyzing data"* (Agresti & Finlay, 1997)
- Statistical methods can be used to find answers to the questions like:
  - What kind and how much data need to be collected?
  - How should we organize and summarize the data?
  - How can we analyze the data and draw conclusions from it?
  - How can we assess the strength of the conclusions and evaluate their uncertainty?



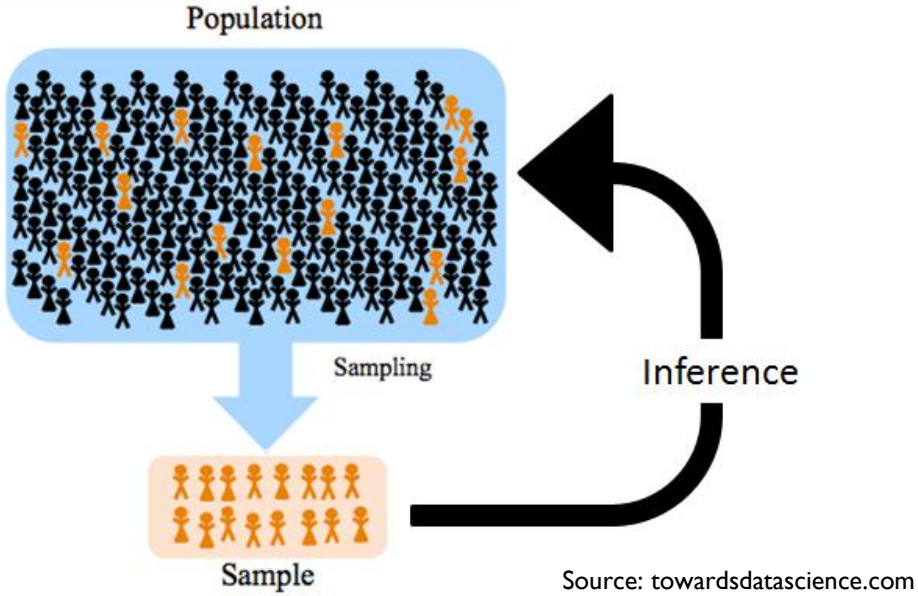
Introduction

- That is, statistics provides methods for
  - Design: Planning and carrying out research studies.
  - Description: Summarizing and exploring data.
  - Inference: Making predictions and generalizing about phenomena represented by the data.



Population and Sample

- "*Population is the collection of all individuals or items under consideration in a statistical study*" (Weiss, 1999)
- *"Sample is that part of the population from which information is collected."* (Weiss, 1999)



Source: towardsdatascience.com

Population and Sample

- Always only a certain, relatively few, features of individual person or object are under investigation at the same time. Not all the properties are wanted to be measured from individuals in the population.
- "*A (statistical) population is the set of measurements (or record of some qualitative trait) corresponding to the entire collection of units for which inferences are to be made.*" (Johnson & Bhattacharyya,1992)
- *"A sample from statistical population is the set of measurements that are actually collected in the course of an investigation."* (Johnson & Bhattacharyya, 1992)



Descriptive and Inferential Statistics

- *"**Descriptive statistics** consist of methods for organizing and summarizing information"* (Weiss, 1999)
  - includes the construction of graphs, charts, and tables, and the calculation of various descriptive measures such as averages, measures of variation, and percentiles.

- *"**Inferential statistics** consist of methods for drawing and measuring the reliability of conclusions about population based on information obtained from a sample of the population."* (Weiss, 1999)
  - includes methods like point estimation, interval estimation and hypothesis testing which are all based on probability theory.

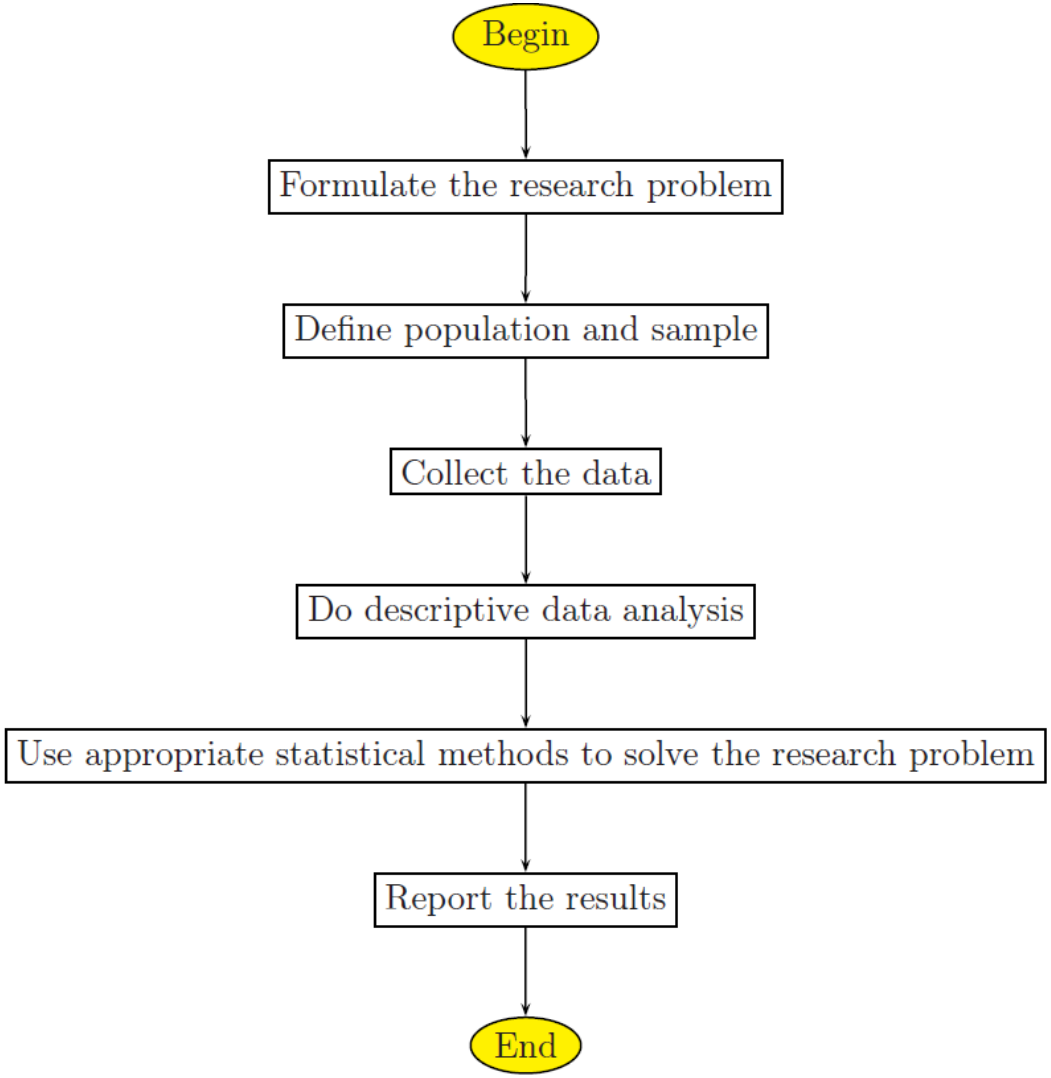


Parameters and Statistics

- *"A parameter is an unknown numerical summary of the population. A statistic is a known numerical summary of the sample which can be used to make inference about parameters."* (Agresti & Finlay, 1997)
- Usually the features of the population under investigation can be summarized by *numerical parameters*. Hence the research problem usually becomes as on investigation of the values of parameters.
- These population parameters are unknown and *sample statistics* are used to make inference about them.
- That is, a *statistic describes a characteristic of the sample* which can then be used *to make inference about unknown parameters*.



Statistical data analysis



Variables and organization of the data

- A characteristic that varies from one person or thing to another is called a *variable*, i.e., a variable is any characteristic that varies from one individual member of the population to another.
  - Examples of variables for humans are height, weight, sex, marital status, and eye color.
  - The first two of these variables yield numerical information (yield numerical measurements) and are examples of quantitative (or numerical) variables, last three yield non-numerical information (yield non-numerical measurements) and are examples of qualitative (or categorical) variables.
  - Quantitative variables can be classified as either *discrete or continuous*.
    - A discrete variable is a variable whose possible values are some or all of the ordinary counting numbers like 0, 1, 2, 3, ...
    - Continuous variables. Quantities such as length, weight, or temperature can in principle be measured arbitrarily accurately



Variables and organization of the data

- Observing the values of the variables for one or more people or things yield data. Each individual piece of data is called an observation and the collection of all observations for particular variables is called a data set or data matrix.

$$\begin{array}{c} \\ \text{Sampling units} \end{array} \begin{array}{c} \text{Variables} \\ \begin{pmatrix} x_{11} & x_{12} & x_{13} & \dots & x_{1k} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2k} \\ x_{31} & x_{32} & x_{33} & \dots & x_{3k} \\ & \vdots & & \ddots & \\ x_{n1} & x_{n2} & x_{n3} & \dots & x_{nk} \end{pmatrix} \end{array}$$

where x_{ij} is a value of the j:th variable collected from i:th observation, $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, k$.



Describing data by tables and graphs

Qualitative variable

- The *number of observations* that fall into particular class (or category) of the qualitative variable is called the frequency (or count) of that class. A table listing all *classes and their frequencies* is called a frequency distribution.

$$\text{Relative frequency of the class} = \frac{\text{Frequency in the class}}{\text{Total number of observation}}$$

- A table listing all classes and their relative frequencies is called a relative frequency distribution. The relative frequencies provide the most relevant information as to the pattern of the data.
- A cumulative frequency (cumulative relative frequency) is obtained by summing the frequencies (relative frequencies) of all classes up to the specific class.
- The qualitative data are presented graphically either as a pie chart or as a horizontal or vertical bar graph.



Describing data by tables and graphs

EXAMPLE 3.1. Let the blood types of 40 persons are as follows:

O O A B A O A A A O B O B O O A O O A A A A AB A B A A O O
O O A A A O A O O AB

Table 1: Frequency distribution of blood types

BLOOD

BLOOD		Statistics	
		Frequency	Percent
Valid	O	16	40.0
	A	18	45.0
	B	4	10.0
	AB	2	5.0
	Total	40	100.0

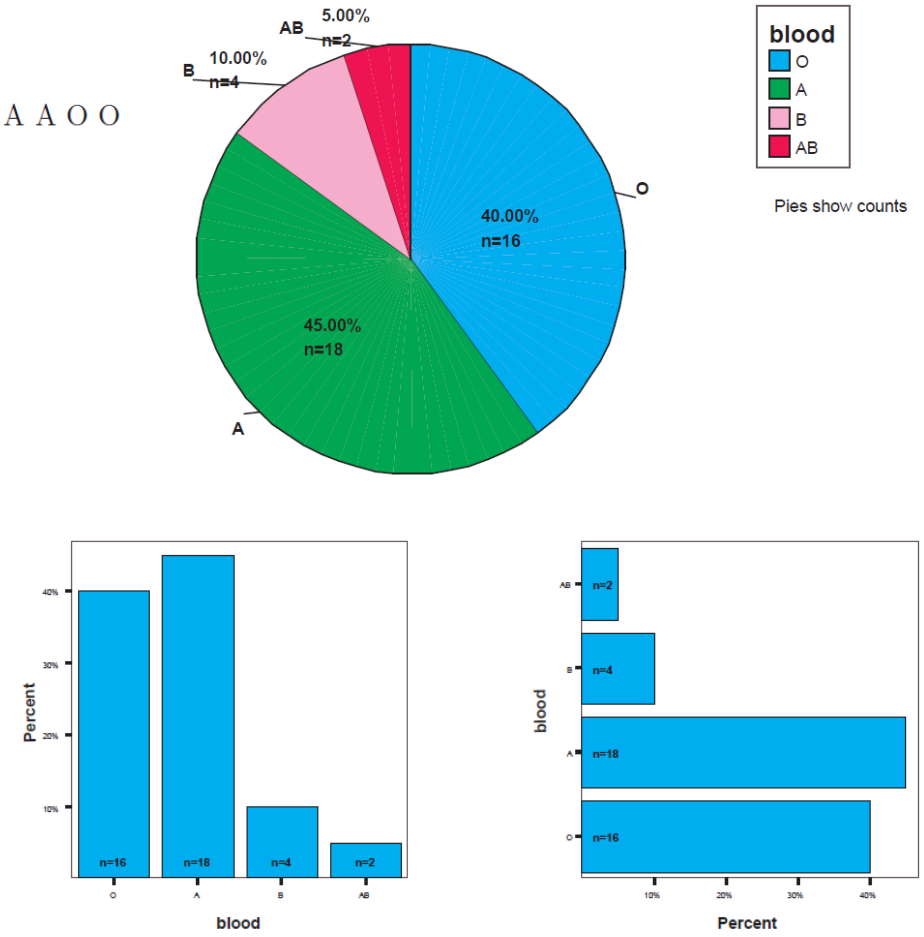


Figure 2: Charts for blood types

Describing data by tables and graphs

Quantitative variable

- If the discrete variable can obtain only few different values, then the data of the discrete variable can be summarized in a same way as qualitative variables in a frequency table.
- If the discrete variable can have a lot of different values or the quantitative variable is the continuous variable, then the data must be grouped into classes (categories) before the table of frequencies can be formed.
- The main steps in a process of grouping quantitative variable into classes are:
  - Find the minimum and the maximum values variable have in the data set
  - Choose intervals of equal length that cover the range between the minimum and the maximum without overlapping. These are called class intervals, and their end points are called class limits.
  - Count the number of observations in the data that belongs to each class interval. The count in each class is the class frequency.
  - Calculate the relative frequencies of each class by dividing the class frequency by the total number of observations in the data.



Describing data by tables and graphs

EXAMPLE 3.2. Age (in years) of 102 people:

34,67,40,72,37,33,42,62,49,32,52,40,31,19,68,55,57,54,37,32,
54,38,20,50,56,48,35,52,29,56,68,65,45,44,54,39,29,56,43,42,
22,30,26,20,48,29,34,27,40,28,45,21,42,38,29,26,62,35,28,24,
44,46,39,29,27,40,22,38,42,39,26,48,39,25,34,56,31,60,32,24,
51,69,28,27,38,56,36,25,46,50,36,58,39,57,55,42,49,38,49,36,
48.44

Table 2: Frequency distribution of people's age

Frequency distribution of people's age

		Frequency	Percent	Cumulative Percent
Valid	18 - 22	6	5.9	5.9
	23 - 27	10	9.8	15.7
	28 - 32	14	13.7	29.4
	33 - 37	11	10.8	40.2
	38 - 42	19	18.6	58.8
	43 - 47	8	7.8	66.7
	48 - 52	12	11.8	78.4
	53 - 57	12	11.8	90.2
	58 - 62	4	3.9	94.1
	63 - 67	2	2.0	96.1
	68 - 72	4	3.9	100.0
	Total	102	100.0	

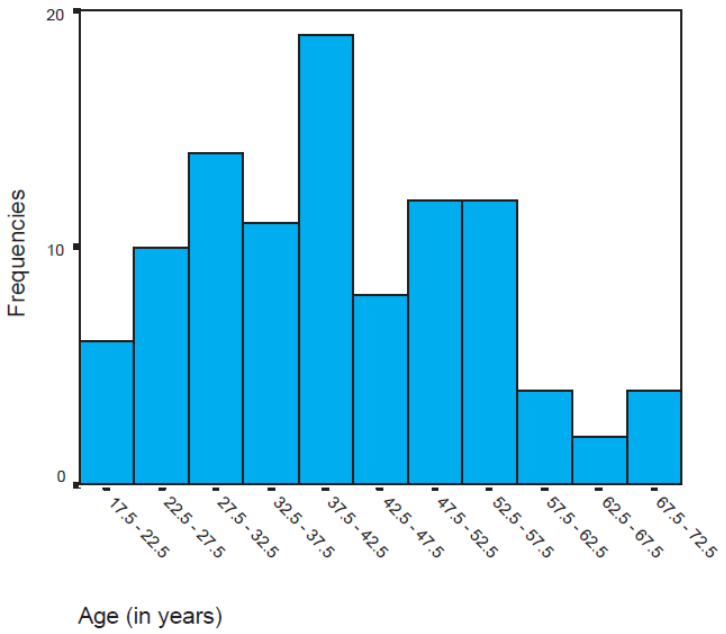


Figure 3: Histogram for people's age

Sample and Population Distributions

- Frequency distributions for a variable apply both to a *population* and to *samples from that population*.
- The first type is called the *population distribution* of the variable, and the second type is called *a sample distribution*.
  - In a sense, the sample distribution is a blurry photograph of the population distribution.
- As the sample size increases, the *sample relative frequency* in any class interval gets closer to the true population relative frequency.
  - Thus, the photograph gets clearer, and the sample distribution looks more like the population distribution.



Sample and Population Distributions

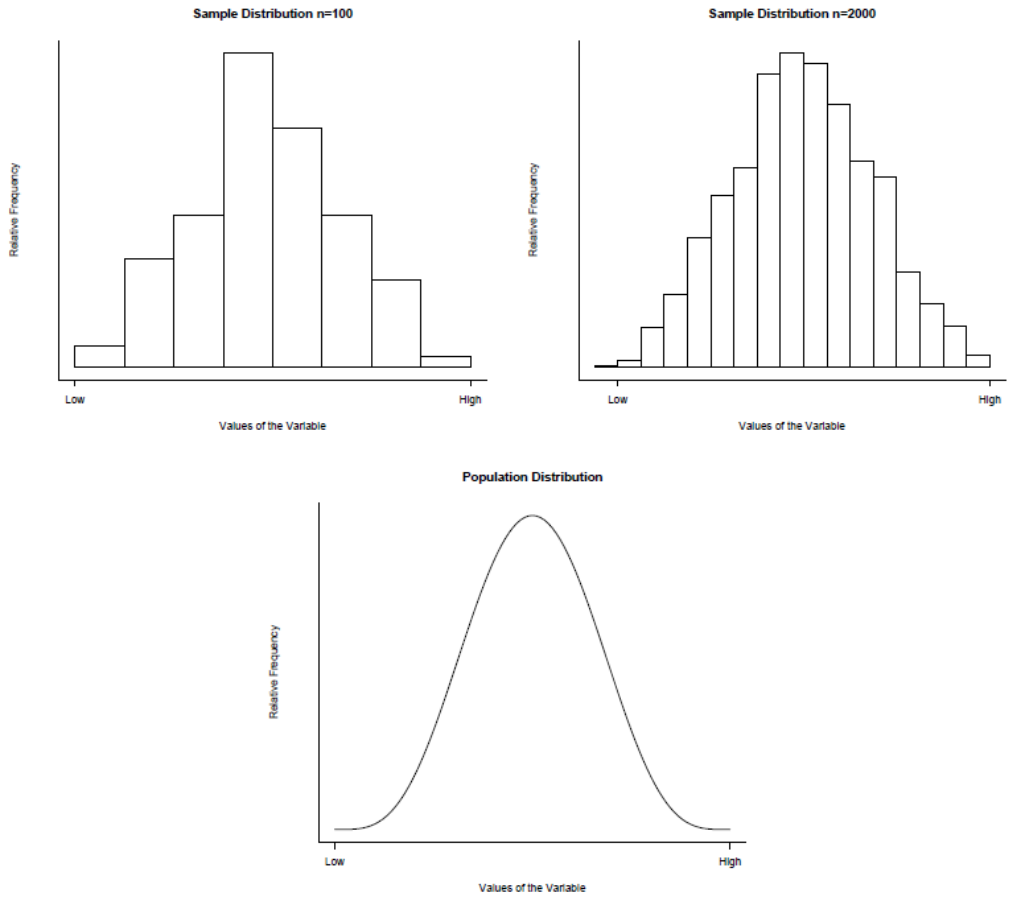


Figure 6: Sample and Population Distributions

Sample and Population Distributions

- One way to summarize a sample of population distribution is to describe its shape.
- The bell-shaped and U-shaped distributions are symmetric.
- On the other hand, a non-symmetric distribution is said to be skewed to the right or skewed to the left, according to which tail is longer.

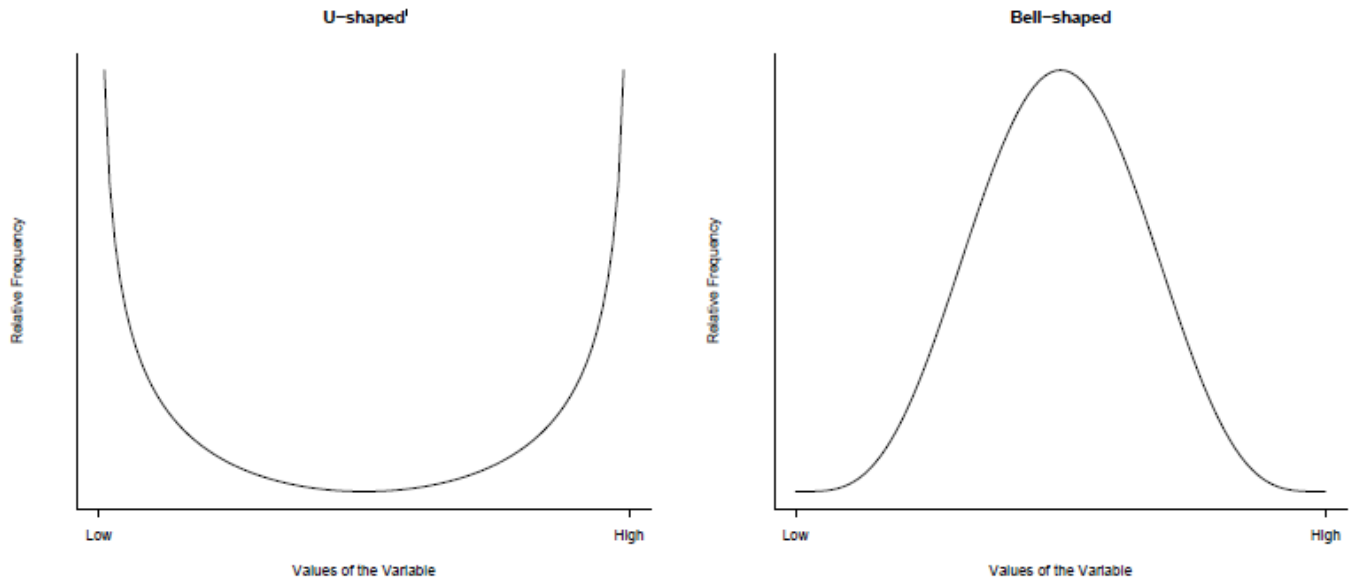


Figure 7: U-shaped and Bell-shaped Frequency Distributions

Sample and Population Distributions

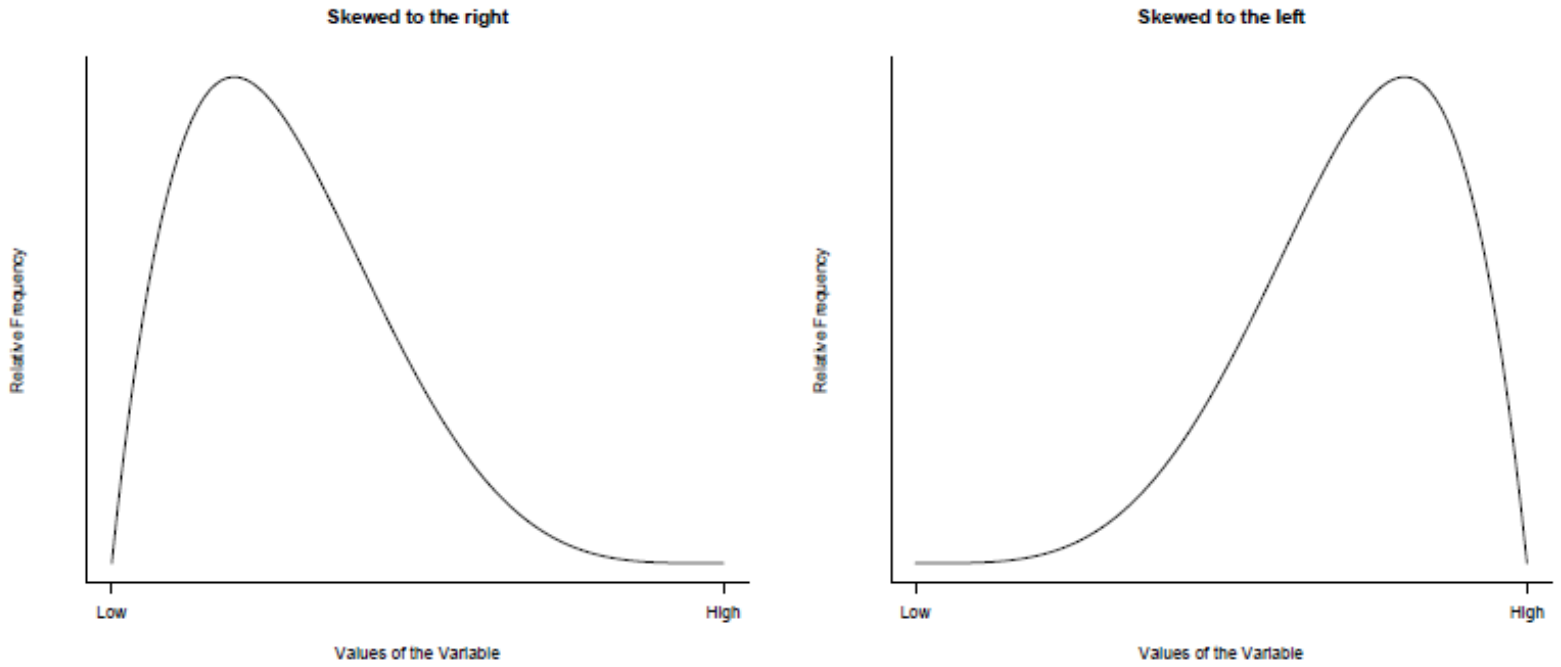


Figure 8: Skewed Frequency Distributions

Measures of center

- *Descriptive measures* that indicate where the center or the most typical value of the variable lies in collected set of measurements are called measures of center.
- Mode
  - The sample mode of a qualitative or a discrete quantitative variable is that *value of the variable which occurs with the greatest frequency* in a data set.
  - Obtain the frequency of each observed value of the variable in a data and note the greatest frequency.
    - If the greatest frequency is one (i.e. no value occurs more than once), then the variable has no mode.
    - If the greatest frequency is 2 or greater, then any value that occurs with that greatest frequency is called a sample mode of the variable.



Measures of center

Table 6: Frequency distribution of blood type

BLOOD

BLOOD		Statistics	
		Frequency	Percent
Valid	O	16	40.0
	A	18	45.0
	B	4	10.0
	AB	2	5.0
	Total	40	100.0

We can see from frequency table that the mode of blood types is A.



Measures of center

- Median
  - Arrange the observed values of variable in a data in increasing order.
    - If the number of observation is odd, then the sample median is the observed value exactly in the middle of the ordered list.
    - If the number of observation is even, then the sample median is the number halfway between the two middle observed values in the ordered list.
  - In both cases, if we let *n* denote the number of observations in a data set, then the sample median is at position $(n+1)/2$ in the ordered list.
  - Example: 7 participants in bike race had the following finishing times in minutes: 28,22,26,29,21,23,24.
    - What is the median?
  - The median is a "central" value – there are as many values greater than it as there are less than it.



Measures of center

- The most commonly used measure of center for quantitative variable is the (arithmetic) sample mean.
- The sample mean of the variable is the *sum of observed values in a data divided by the number of observations*.

The sample mean *of the variable is the sum of observed values* $x_1, x_2, x_3, \ldots, x_n$ *in a data divided by the number of observations* n. *The sample mean is denoted by* $\bar{x}$, *and expressed operationally,*

If the sample size is n, then the mean of the variable x is

$$\frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}.$$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \quad \text{or} \quad \frac{\sum x_i}{n}.$$



Measures of center

- Which measure to choose?
  - The mode should be used when calculating measure of center for the qualitative variable.
  - When the variable is quantitative with symmetric distribution, then the mean is proper measure of center.
  - In a case of quantitative variable with skewed distribution, the median is good choice for the measure of center.
    - This is related to the fact that the mean can be highly influenced by an observation that falls far from the rest of the data, *called an outlier.*
- If population mode, the population median and the population mean, which are all unknown. Then the sample mode, the sample median and the sample mean can be used *to estimate* the values of these corresponding unknown population values.



Measures of variation

- To measure the extent of variation around the center.
  - Two data sets of the same variable may exhibit similar positions of center but may be remarkably different with respect to variability.
- Measures of variation are used **mostly** only for quantitative variables.
- Range:
  - The sample range of the variable is the difference between its maximum and minimum values in a data set: *Range* = *Max* −*Min*.
    - Quite easy to compute but a great deal of information is ignored, that is, only the largest and smallest values of the variable are considered; the other observed values are disregarded.



Measures of variation

- The median of the variable divides the observed values into two equal parts – the bottom 50% and the top 50%.
- The percentiles of the variable divide observed values into hundredths, or 100 equal parts.
  - Roughly speaking, the first percentile, P1, is the number that divides the bottom 1% of the observed values from the top 99%; second percentile, P2, is the number that divides the bottom 2% of the observed values from the top 98%; and so forth. The median is the 50th percentile.
- The deciles of the variable divide the observed values into tenths, or 10 equal parts.
  - The variable has nine deciles, denoted by D1, D2, . . . , D9. The first decile D1 is 10th percentile, the second decile D2 is the 20th percentile, and so forth.
- The most commonly used percentiles are quartiles. The quartiles of the variable divide the observed values into quarters, or 4 equal parts.



Measures of variation

- The variable has three quartiles, denoted by *Q1*, *Q2* and *Q3*.
  - the first quartile, *Q1*, is the number that divides the bottom 25% of the observed values from the top 75%;
  - second quartile, *Q2*, is the median, which is the number that divides the bottom 50% of the observed values from the top 50%; and
  - the third quartile, *Q3*, is the number that divides the bottom 75% of the observed values from the top 25%.



Measures of variation

- Let *n* denote the number of observations in a data set. Arrange the observed values of variable in a data in increasing order.
  - The first quartile *Q1* is at position *(n+1)/4* ,
  - The second quartile *Q2* (the median) is at position *(n+1)/2* ,
  - The third quartile *Q3* is at position *3(n+1)/4* ,

  in the ordered list.
- If a position is not a whole number, linear interpolation is used.
- ***Inter-quartile range (IQR):*** The sample IQR is the difference between the first and third quartiles of the variable, that is, *IQR = Q3 − Q1*.
  - Roughly speaking, the IQR gives the range of the middle 50% of the observed values.



Measures of variation

Five-number summary and boxplots

- Minimum, maximum and quartiles together provide information on center and variation of the variable in a nice compact way.
- The five-number summary of the variable consists of minimum, maximum, and quartiles written in increasing order: *Min, Q1, Q2, Q3, Max*.
- A boxplot is based on the five-number summary and can be used to provide a graphical display of the center and variation of the observed values of variable in a data set.
- Actually, two types of boxplots are in common use –boxplot and modified boxplot.
  - The difference is that potential outliers (i.e. observed value, which do not appear to follow the characteristic distribution of the rest of the data) are plotted individually in a modified boxplot, but not in a boxplot.



Measures of variation

- To construct a boxplot
  - Determine the five-number summary
  - Draw a horizontal (or vertical) axis on which the numbers obtained in step 1 can be located. Above this axis, mark the quartiles and the minimum and maximum with vertical (horizontal) lines.
  - Connect the quartiles to each other to make a box, and then connect the box to the minimum and maximum with lines.

Five-number summary

Price ($/oz)

N	Valid	54
	Missing	0
Median		.1000
Minimum		.05
Maximum		.25
Percentiles	25	.0700
	50	.1000
	75	.1325

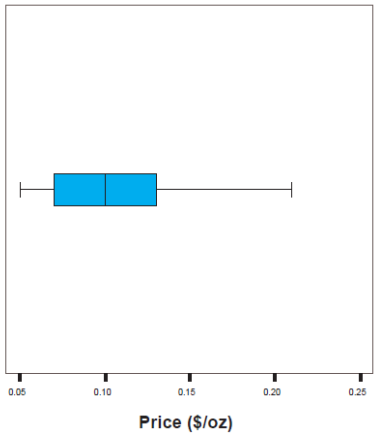


Figure 9: Boxplot for the prices of hotdogs



Measures of variation

Standard deviation

- It can be considered as a kind of average of the absolute deviations of observed values from the mean of the variable in question.
- For a variable *x*, the sample standard deviation, denoted by s_x is

$$s_x = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}.$$

The standard deviation is always positive number, i.e., $s_x > 0$.



Measures of variation

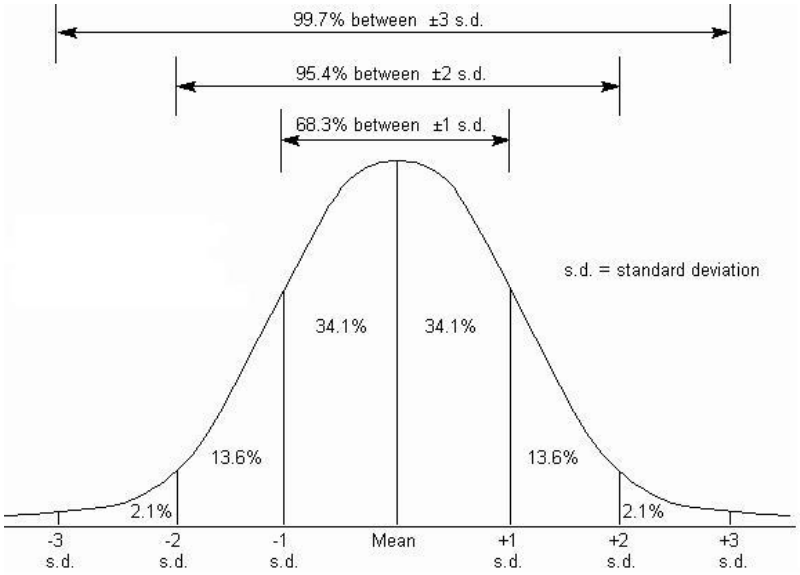
Empirical rule for symmetric distributions

- For bell-shaped symmetric distributions (like the normal distribution), empirical rule relates the standard deviation to the proportion of the observed values of the variable in a data set that lie in a interval around the mear $\bar{x}$.
- Empirical guideline for symmetric bell-shaped distribution, approximately

68% of the values lie within $\bar{x} \pm s_x$,

95% of the values lie within $\bar{x} \pm 2s_x$,

99.7% of the values lie within $\bar{x} \pm 3s_x$.



Measures of variation

Sample statistics and population parameters

- A statistic describes a sample, while a parameter describes the population from which the sample was taken.
- We call μ and σ the population mean and population standard deviation.
- Whereas the statistics $\bar{x}$ are s variables, with values depending on the sample chosen, the parameters μ and σ are constants.
- we make inferences about unknown parameters (such as μ and σ) using sample statistics (such as $\bar{x}$ and s).
- Inferential statistical methods use sample data to make predictions about the values of useful summary descriptions, called parameters, of the population of interest.



Probability Distributions

Probability distributions

- The probability of a particular outcome is the proportion of times that outcome would occur in a long run of repeated observations.
- A simplified representation of such an experiment is a *very long sequence of flips* of a coin, the outcome of interest being that a head faces upwards. Any on flip may or may not result in a head. If the coin is balanced, then a basic result in probability, called *law of large numbers*, implies that the *proportion of flips resulting in a head tends toward ½ as the number of flips increases*. Thus, the probability of a head in any single flip of the coin equals ½.



Probability Distributions

- A variable which can take at least two different numerical values in a long run of repeated observations is called **random variable**.
- *"A random variable is a variable whose value is a numerical outcome of a random phenomenon."*
- The **probability distribution** of the random variable *X* lists the possible outcomes together with their probabilities the variable *X* can have.

The probability distribution of a *discrete* random variable X assigns a probability to each possible values of the variable. Each probability is a number between 0 and 1, and the sum of the probabilities of all possible values equals 1. Let x_i, $i = 1, 2, \ldots, k$, denote a possible outcome for the random variable X, and let $P(X = x_i) \models P(x_i) = p_i$ denote the probability of that outcome. Then

$$0 \leq P(x_i) \leq 1 \quad \text{and} \quad \sum_{i=1}^{k} P(x_i) = 1$$

since each probability falls between 0 and 1, and since the total probability equals 1.



Probability Distributions

DEFINITION 6.3 (Probability distribution of a discrete random variable). *A discrete random variable X has a countable number of possible values. The probability distribution of X lists the values and their probabilities:*

Value of X	x_1	x_2	x_3	$\ldots$	x_k
Probability	$P(x_1)$	$P(x_2)$	$P(x_3)$	$\ldots$	$P(x_k)$

The probabilities $P(x_i)$ must satisfy two requirements:

1. *Every probability $P(x_i)$ is a number between 0 and 1.*
2. $P(x_1) + P(x_2) + \cdots + P(x_k) = 1.$



Probability Distributions

- We can use a **probability histogram** to picture the probability distribution of a discrete random variable.
- Furthermore, we can find the probability of any event [such as $P(X \le x_i)$ or $P(x_i \le X \le x_j), i \le j$] by adding the probabilities $P(x_i)$ of the particular values x_i that make up the event.

Probability distribution of a continuous random variable

- A continuous random variable X takes all values in an interval of numbers [a, b]. The probability distribution of X describes the probabilities $P(x_1 \le X \le x_2)$ of all possible intervals of numbers $[x_1, x_2]$.

  The probabilities $P(x_1 \le X \le x_2)$ must satisfy two requirements:

  - For every interval $[x_1, x_2]$, the probability $P(x_1 \le X \le x_2)$ is a number between 0 and 1.
  - $P(a \le X \le b) = 1$.



Probability Distributions

- The probability model for a continuous random variable assign probabilities to intervals of outcomes rather than to individual outcomes.
- The probability distribution of a continuous random variable is pictured by a **density curve**. A density curve is smooth continuous curve having area exactly one.
- *In fact, the population distribution of a variable is, equivalently, the probability distribution for the value of that variable for a subject selected randomly from the population.*



Probability Distributions

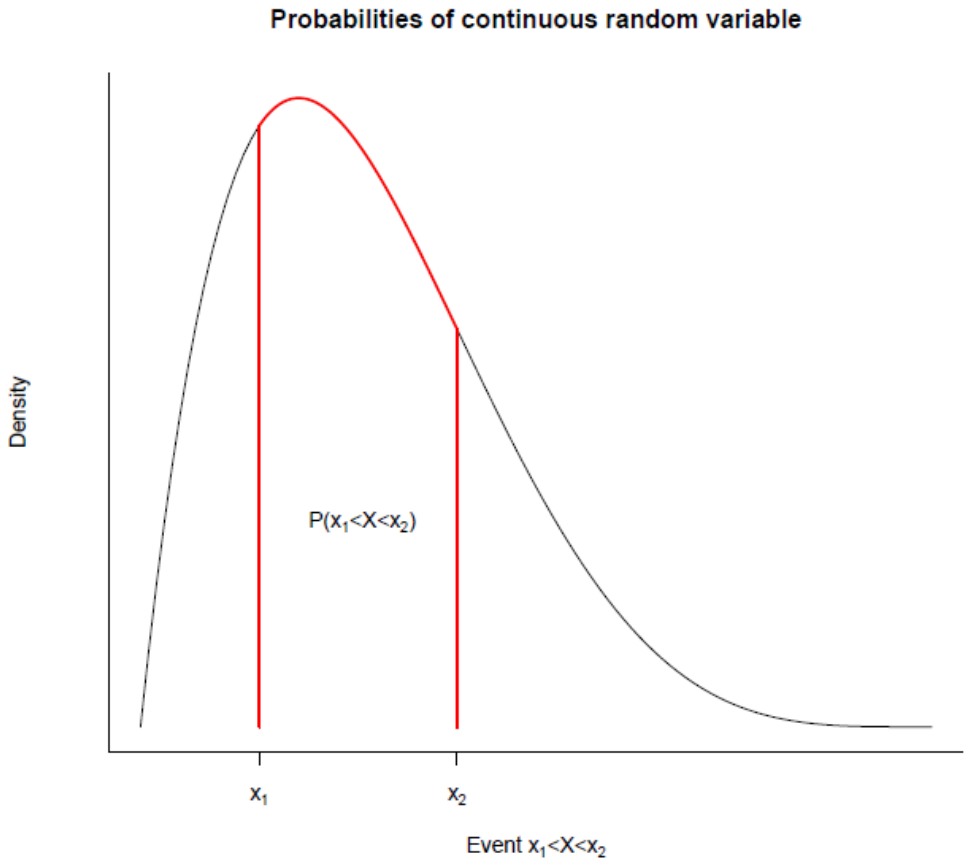


Figure 10: The probability distribution of a continous random variable assign probabilities as areas under a density curve.

Mean and standard deviation of random variable

DEFINITION 6.5 (Mean of a discrete random variable). *Suppose that X is a discrete random variable whose probability distribution is*

Value of X	x_1	x_2	x_3	$\dots$	x_k
Probability	$P(x_1)$	$P(x_2)$	$P(x_3)$	$\dots$	$P(x_k)$

The mean of the discrete random variable X is

$$\begin{aligned}\mu &= x_1P(x_1) + x_2P(x_2) + x_3P(x_3) + \cdots + x_kP(x_k)\\ &= \sum_{i=1}^{k} x_iP(x_i).\end{aligned}$$

The mean μ is also called the ***expected value*** *of X and is denoted by $E(X)$.*



Mean and standard deviation of random variable

DEFINITION 6.6 (Standard deviation of a discrete random variable). *Suppose that X is a discrete random variable whose probability distribution is*

Value of X	x_1	x_2	x_3	$\dots$	x_k
Probability	$P(x_1)$	$P(x_2)$	$P(x_3)$	$\dots$	$P(x_k)$

and that μ is the mean of X. The ***variance*** *of the discrete random variable X is*

$$\sigma^2 = (x_1 - \mu)^2 P(x_1) + (x_2 - \mu)^2 P(x_2) + (x_3 - \mu)^2 P(x_3) + \cdots + (x_k - \mu)^2 P(x_k)$$
$$= \sum_{i=1}^{k} (x_i - \mu)^2 P(x_i).$$

The standard deviation σ of X is the square root of the variance.



Normal distribution

- A continuous random variable graphically described by a certain bell-shaped density curve is said to have the normal distribution.
- A continuous random variable X following normal distribution has two parameters: the mean μ and the standard deviation σ .

DEFINITION 6.7 (Normal distribution). *A continuous random variable X is said to be normally distributed or to have a normal distribution if its density curve is a symmetric, bell-shaped curve, characterized by its mean μ and standard deviation σ. For each fixed number z, the probability concentrated within interval $[\mu - z\sigma, \mu + z\sigma]$ is the same for all normal distributions. Particularly, the probabilities*

$$P(\mu - \sigma < X < \mu + \sigma) = 0.683 \quad (4)$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.954 \quad (5)$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.997 \quad (6)$$

hold. A random variable X following normal distribution with a mean of μ and a standard deviation of σ is denoted by $X \sim N(\mu, \sigma)$.



Normal distribution

The normal density curves are specified by a particular equation. The height of the density curve at any point x is given by the **density function**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}. \tag{7}$$

We will not make direct use of this fact, although it is the basis of mathematical work with normal distribution. Note that the density function is completely determined by μ and σ.

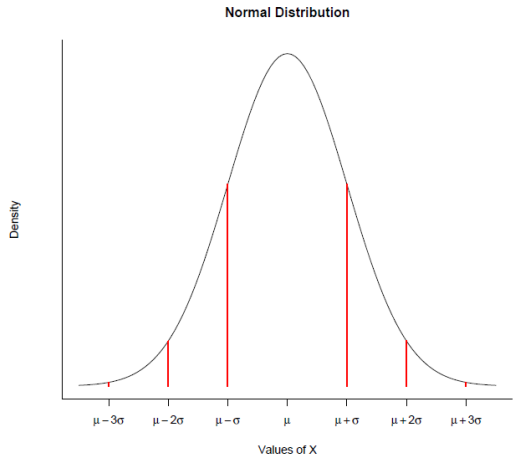


Figure 11: Normal distribution.

Normal distribution

DEFINITION 6.8 (Standard normal distribution). *A continuous random variable Z is said to have a standard normal distribution if Z is normally distributed with mean $\mu = 0$ and standard deviation $\sigma = 1$, i.e., $Z \sim N(0, 1)$.*

The **standard normal table** can be used to calculate probabilities concerning the random variable Z. The standard normal table gives area to the left of a specified value of z under density curve:

$$P(Z \leq z) = \text{Area under curve to the left of } z.$$

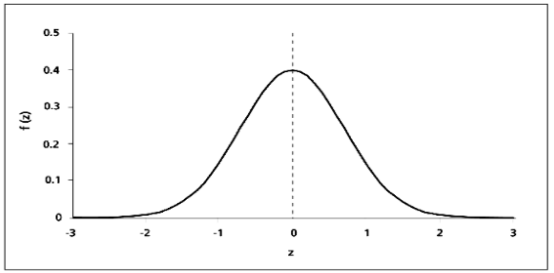
For the probability of an interval $[a, b]$:

$$P(a \leq Z \leq b) = [\text{Area to left of } b] - [\text{Area to left of } a].$$

The following properties can be observed from the symmetry of the standard normal distribution about 0:

(a) $P(Z \leq 0) = 0.5$,

(b) $P(Z \leq -z) = 1 - P(Z \leq z) = P(Z \geq z)$.



Characteristics of the Normal distribution

- Symmetric, bell shaped
- Continuous for all values of X between $-\infty$ and ∞ so that each conceivable interval of real numbers has a probability other than zero.
- $-\infty \leq X \leq \infty$
- Two parameters, μ and σ. Note that the normal distribution is actually a family of distributions, since μ and σ determine the shape of the distribution.

- The rule for a normal density function is

$$f(x;\, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

- The notation $N(\mu, \sigma^2)$ means normally distributed with mean μ and variance σ^2. If we say $X \sim N(\mu, \sigma^2)$ we mean that X is distributed $N(\mu, \sigma^2)$.
- About 2/3 of all cases fall within one standard deviation of the mean, that is

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = .6826.$$

- About 95% of cases lie within 2 standard deviations of the mean, that is

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = .9544$$



Why is the normal distribution useful?

- **Many things actually are normally distributed, or very close to it.** For example, height and intelligence are approximately normally distributed; measurement errors also often have a normal distribution
- **The normal distribution is easy to work with mathematically**. In many practical cases, the methods developed using normal theory work quite well even when the distribution is not normal.
- There is a very strong connection between the size of a sample N and the extent to which a sampling distribution approaches the normal form. Many sampling distributions based on large N can be approximated by the normal distribution even though the population distribution itself is definitely not normal.



The standardized normal distribution

a. General Procedure. As you might suspect from the formula for the normal density function, it would be difficult and tedious to do the calculus every time we had a new set of parameters for μ and σ. So instead, we usually work with the standardized normal distribution, where $\mu = 0$ and $\sigma = 1$, i.e. N(0,1). That is, rather than directly solve a problem involving a normally distributed variable X with mean μ and standard deviation σ, an indirect approach is used.

1. We first convert the problem into an equivalent one dealing with a normal variable measured in standardized deviation units, called a standardized normal variable. To do this, if $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$$

2. A table of standardized normal values (Appendix E, Table I) can then be used to obtain an answer in terms of the converted problem. StatistialTables.pdf

3. If necessary, we can then convert back to the original units of measurement. To do this, simply note that, if we take the formula for Z, multiply both sides by σ, and then add μ to both sides, we get

$$X = Z\sigma + \mu$$

4. The interpetation of Z values is straightforward. Since $\sigma = 1$, if $Z = 2$, the corresponding X value is exactly 2 standard deviations above the mean. If $Z = -1$, the corresponding X value is one standard deviation below the mean. If $Z = 0$, X = the mean, i.e. μ.

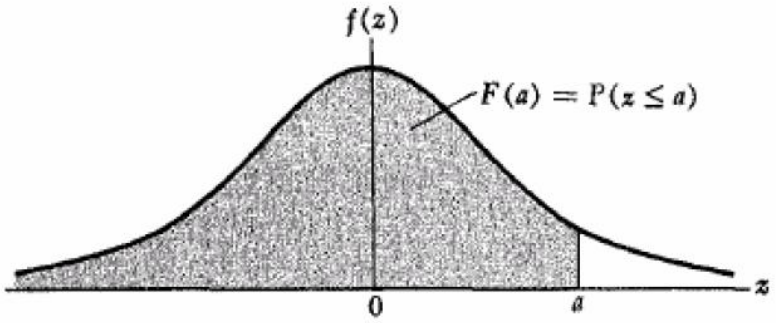


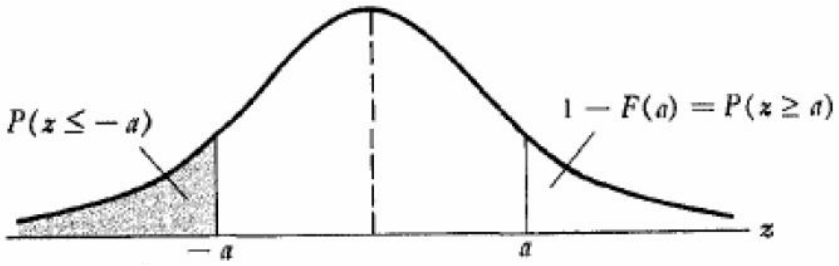
b. Rules for using the standardized normal distribution. It is very important to understand how the standardized normal distribution works, so we will spend some time here going over it. Recall that, for a random variable X,

1. **$P(Z \le a)$**
   $= F(a)$ (use when a is positive)
   $= 1 - F(-a)$ (use when a is negative)

Find $P(Z \le a)$ for a = 1.65, -1.65, 1.0, -1.0

$P(Z \le 1.65) = F(1.65) = .95$
$P(Z \le -1.65) = F(-1.65) = 1 - F(1.65) = .05$
$P(Z \le 1.0) = F(1.0) = .84$
$P(Z \le -1.0) = F(-1.0) = 1 - F(1.0) = .16$



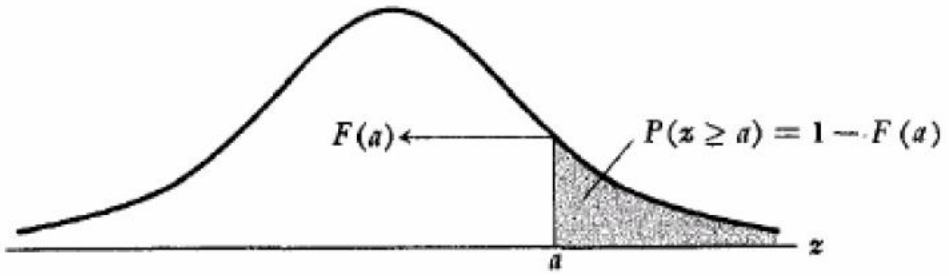


2. **$P(Z \geq a)$**
$= 1 - F(a)$ (use when a is positive)
$= F(-a)$ (use when a is negative)

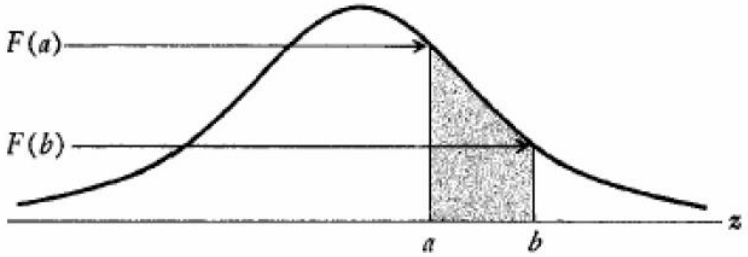
Find $P(Z \geq a)$ for a = 1.5, -1.5

$P(Z \geq 1.5) = 1 - F(1.5) = 1 - .9332 = .0668$

$P(Z \geq -1.5) = F(1.5) = .9332$



3. **$P(a \leq Z \leq b) = F(b) - F(a)$**



Find $P(a \leq Z \leq b)$ for a = -1 and b = 1.5

$P(-1 \leq Z \leq 1.5) = F(1.5) - F(-1) = F(1.5) - (1 - F(1)) = .9332 - 1 + .8413 = .7745$



4. For a positive, $P(-a \le Z \le a) = 2F(a) - 1$

PROOF:

$P(-a \le Z \le a)$
$= F(a) - F(-a)$ (by rule 3)
$= F(a) - (1 - F(a))$ (by rule 1)
$= F(a) - 1 + F(a)$
$= 2F(a) - 1$

EX: find $P(-a \le Z \le a)$ for $a = 1.96$, $a = 2.58$

$P(-1.96 \le Z \le 1.96) = 2F(1.96) - 1 = (2 * .975) - 1 = .95$
$P(-2.58 \le Z \le 2.58) = 2F(2.58) - 1 = (2 * .995) - 1 = .99$

1. The top 5% of applicants (as measured by GRE scores) will receive scholarships. If GRE ~ $N(500, 100^2)$, how high does your GRE score have to be to qualify for a scholarship?



1. The top 5% of applicants (as measured by GRE scores) will receive scholarships. If GRE ~ $N(500, 100^2)$, how high does your GRE score have to be to qualify for a scholarship?

Solution. Let X = GRE. We want to find x such that

$$P(X \geq x) = .05$$

This is too hard to solve as it stands - so instead, compute

$$Z = (X - 500)/100 \qquad \text{(NOTE: } Z \sim N(0,1) \text{)}$$

and find z for the problem,

$$P(Z \geq z) = .05$$

Note that $P(Z \geq z) = 1 - F(z)$ (Rule 2). If $1 - F(z) = .05$, then $F(z) = .95$. Looking at Table I in Appx E, $F(z) = .95$ for $z = 1.65$ (approximately).

Hence, $z = 1.65$. To find the equivalent x, compute

$x = (z * 100) + 500 = (1.65 * 100) + 500 =$ 665. Thus, your GRE score needs to be 665 or higher to qualify for a scholarship.



2. Family income ~ N($25000, $10000^2). If the poverty level is $10,000, what percentage of the population lives in poverty?



2. Family income ~ N($25000, $10000^2). If the poverty level is $10,000, what percentage of the population lives in poverty?

Solution. Let X = Family income. We want to find P(X ≤ $10,000). This is too hard to compute directly, so let

Z = (X - $25,000)/$10,000.

If x = $10,000, then z = ($10,000 - $25,000)/$10,000 = -1.5. So,

P(X ≤ $10,000) = P(Z ≤ -1.5) = F(-1.5) = 1 - F(1.5) = 1 - .9332 = .0668. Hence, a little under 7% of the population lives in poverty.



3. A new tax law is expected to benefit "middle income" families, those with incomes between \$20,000 and \$30,000. If Family income ~ N(\$25000, $\$10000^2$), what percentage of the population will benefit from the law?



3. A new tax law is expected to benefit "middle income" families, those with incomes between \$20,000 and \$30,000. If Family income ~ N(\$25000, $\$10000^2$), what percentage of the population will benefit from the law?

Solution. Let X = Family income. We want to find P(\$20,000 X ≤ \$30,000). To solve, let

$$Z = (X - \$25{,}000)/\$10{,}000.$$

Note that when x = \$20,000, z = (\$20,000 - \$25,000)/\$10,000 = -0.5, and when x = \$30,000, z = +0.5. Hence,

$P(\$20{,}000 \le X \le \$30{,}000) = P(-.5 \le Z \le .5) = 2F(.5) - 1 = 1.383 - 1 = .383$. Thus, about 38% of the taxpayers will benefit from the new law.

